```
students = {}

while True:
    show_menu()
    choice = input("Enter your choice (1/2/3/4/5): ")

    if choice == '1':
            add_student(students)
    elif choice == '2':
            view_students(students)
    elif choice == '3':
            update_student(students)
    elif choice == '4':
            delete_student(students)
    elif choice == '5':
            print("Exiting the Student Management System. Goodbye!")
            break
    else:
            print("Invalid choice! Please select a valid option.")
```

In the `main()` function:

- We initialize an empty dictionary `students`.